

Mitchell Tyler Scott

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Curriculum Vitae

CONTACT
INFORMATION Department of Mathematics
Emory University
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EDUCATION **Emory University**, Atlanta, Georgia, USA
Ph.D., Department of Mathematics, expected in May 2027
M.S., Department of Computer Science, expected in August 2026
 Advisor: Professor Yuanzhe Xi
 Tentative thesis title: Data-driven Preconditioners for Data Science Problems
Tufts University, Medford, Massachusetts, USA
M.S., Department of Mathematics, May 2023
 Advisors: Professor Misha Kilmer, Professor Xiaozhe Hu
 Thesis: A Tale of Two Tensors: Using Hierarchical and Block Low Rank Matrices to Make Preconditioners and Save Storage
Cornell University, Ithaca, New York, USA
B.S., Department of Biological Engineering, December 2020
 Minors: Pure Mathematics, Applied Mathematics
 Thesis: Designing and Optimizing a Protocol for Whole-Ovary Vitrification

RESEARCH
INTERESTS Preconditioners for Stochastic Optimization
(randomized) Numerical Linear Algebra
High Performance Scientific Computing
Stochastic Eigen-analysis / Random Matrix Theory

PUBLICATIONS **Journal Articles:**
1. [M.T. Scott](#), T. Xu, Z. Tang, A. Pichette-Emmons, Q. Ye, Y. Saad, and Y. Xi, *Designing Preconditioners for SGD: Local Conditioning, Noise Floors, and Basin Stability*, Transactions of Machine Learning Research, June 2026 [[arXiv](#)] [[openReview](#)] [[video](#)]
In Preparation:
2. J. Haddock, R. Murray, and [M.T. Scott](#), *Block Subset Selection based on Randomized QR with Column Pivoting*
1. [M.T. Scott](#), A. Khan, C. Chen, M. Kilmer, and A. Saibaba, *Tensor-Train Compression of M2L Operators with Direct Application in $\mathcal{H}^2$ Matrix-Vector Products*
Peer Reviewed Teaching Activities:
1. [M.T. Scott](#) (2025) *Markov Chains in Public Transportation*, Teaching Computation in the Sciences Using MATLAB. Submitted October 7, 2025, Accepted October 26, 2025, from <https://serc.carleton.edu/309394>

RESEARCH
EXPERIENCE **Lawrence Livermore National Laboratory** - Livermore, CA
 Visiting Graduate Student Researcher: May 2025 - September 2025
 “Accelerating Multigrid Prolongation via Bipartite Graph Attention Networks”
 Mentors: Rui Peng Li, Socrates Petrides
Lawrence Berkeley National Laboratory - Berkeley, CA

Visiting Graduate Student Researcher: May 2024 - August 2024
 “Quantized Tensor Trains for Solving Maxwell’s Equations with Spectral Methods”
 Mentors: Erika Ye, Chao Yang

CONFERENCES,
WORKSHOPS
AND TALKS

Contributed Talks:

8. *What Makes a Good Data Science Preconditioner?*,
41st Colorado Conference On Iterative and Multigrid Methods, CU Boulder, Boulder, Colorado, June 23, 2026.
7. *What Makes a Good Data Science Preconditioner?*,
Preconditioning techniques for scientific and industrial applications, ICMS, Edinburgh, Scotland, UK, May 27, 2026.
6. *Block Subset Selection based on Randomized QR with Column Pivoting: A $\star_M$ Framework*,
27th Conference of the International Linear Algebra Society, Blacksburg, VA, May 22, 2026.
5. *What Makes a Good Data Science Preconditioner?*,
Southeast Applied and Computational Math Student Workshop, Atlanta, GA, April 3, 2026.
4. *Preconditioning Stochastic Gradient Descent: Theoretical Guarantees & Novel Framework*,
Southeast Applied and Computational Math Student Workshop, Auburn, AL, April 5, 2025.
3. *Preconditioning Stochastic Gradient Descent: Theoretical Guarantees & Novel Framework*,
Atlanta SIAM Student Conference, Atlanta, GA, March 28, 2025.
2. *Constructing Hierarchical Matrices through Recursive Tensor Decomposition*,
Conference on Fast Direct Solvers, West Lafayette, IN, November 4, 2023.
1. *Discovering Hierarchical Matrix Structure Through Recursive Tensor Decomposition: Joint Mathematics Meeting*, Boston, MA, January 4, 2023.

Contributed Posters:

7. *What Makes a Good Preconditioner for Data Science?*. Georgia Scientific Computing Symposium, GSCS 26, Athens, GA, February 21, 2026
6. *What Makes a Good Preconditioner for Data Science?*. Randomized Numerical Linear Algebra Workshop, ICERM, Providence, RI, February 5, 2026.
5. *Accelerating and Stabilizing SGD via Preconditioning: A Unified Analysis*. Learning about Laney Showcase, Emory University, Laney Graduate School, Atlanta, GA, Sept 29, 2025.
4. *Accelerating Multigrid Prolongation via Bipartite Graph Attention Networks*. Summer Student Showcase, Lawrence Livermore National Lab, Livermore, CA, August 6, 2025
3. *Fisher-informed Nonlinear Truncated Generalized Conjugate Residual Optimizer*. Georgia Scientific Computing Symposium, GSCS 25, Atlanta, GA, February 8, 2025
2. *Acceleration Methods for Scientific and Data Science Applications*. SIAM Mathematics of Data Science, SIAM MDS 24, Atlanta, GA, October 25, 2024
1. *Quantized Tensor Trains for Solving Maxwell’s Equations with Spectral Methods*. Computing Sciences Poster Session, Berkeley Lab, Berkeley, CA, August 6, 2024

Student Seminar Talks:

6. *Generalizable Preconditioners*, University of Minnesota Numerical Analysis Seminar, Minneapolis, MN, February 9, 2026.
5. *Random Matrix Theory: A Numericist’s Need-to-know*, University of Minnesota Numerical Analysis Seminar, Minneapolis, MN, November 6, 2025.
4. *All Aboard the Tensor Train: Using Spectral Methods in the Tensor Train Framework to solve Maxwell’s Equation*, Tufts University SIAM Seminar, Medford, MA, February 19, 2025.
3. *Integration over Lie Groups, with applications to random matrix theory*, Emory University

Research in Algebra and Number Theory, Atlanta, GA, April 11, 2024.

2. *A Tale of Two Tensors: Using Hierarchical and Block Low Rank to Make Preconditioners and Save Storage*, Emory University Discussions in Scientific Computing, Atlanta, GA, October 6, 2023.
1. *It's Tensor Time!: A Computational Framework for Analyzing Structured Matrices*, Tufts University Monday Math Meeting, Medford, MA, September 12, 2022.

Workshops and Conferences Attended

6. *Preconditioning techniques for scientific and industrial applications*, Precond 26, ICMS, Bayes Centre, Edinburgh Scotland, UK, May 27-29, 2026.
5. *27th Conference of the International Linear Algebra Society ILAS 2026*, Virginia Tech, Blacksburg, VA, May 18-22, 2026.
4. *Randomized Numerical Linear Algebra*, ICERM, Brown University, Providence, RI, Feb. 2-6, 2026.
3. *Randomized Numerical Linear Algebra Research Collaboration Workshop*, IPAM, Los Angeles, CA, August 10-16, 2025.
2. *Acceleration and Extrapolation Methods*, ICERM, Providence, RI, July 24-28, 2023.
1. *Joint Mathematics Meetings*, Boston, MA, January 4-7, 2023.

TEACHING EXPERIENCE

Emory University

Instructor of Record

- MATH 190 - Mathematics of Sustainability (Spring 2026)
- MATH 111 - Calculus I (Fall 2025)

Teaching Assistant

- MATH 210 - Advanced Calculus for Data Science (Fall 2026): **Head TA**
- MATH 210 - Advanced Calculus for Data Science (Spring 2025): **Head TA**
- MATH 210 - Advanced Calculus for Data Science (Fall 2024)
- MATH 315 - Numerical Analysis (Spring 2024)

Directed Reading Program Mentor

- Introduction to Stochastic Jump Processes (Spring 2026)
- Introduction to Stochastic Processes (Spring 2026)
- Advanced Game Theory: Algorithms and Voting Systems (Fall 2025)
- Introduction to Stochastic Calculus (Spring 2025)
- Introduction to Functional Analysis and Finite Elements (Fall 2024)
- Introduction to Stochastic Processes (Spring 2024)
- Introduction to Iterative Methods for Inverse Problems (Fall 2023)
- Introduction to Stochastic Processes (Fall 2023)

Course Assistant

- MATH 351 - Partial Differential Equations (Fall 2023)

Tufts University

Teaching Assistant

- MATH 126 - Numerical Linear Algebra (Spring 2023)

Directed Reading Program Mentor

- Introduction to Mathematical Control Theory (Fall 2022)

Course Assistant

- MATH 125 - Numerical Analysis (Fall 2022)
- MATH 32 - Calculus I (Spring 2022)

Cornell University

Undergraduate Teaching Assistant

- BEE 2600 - Introduction to Biological Engineering (Fall 2018)

HONORS AND AWARDS

Conference Travel Grant,

Preconditioning techniques for scientific and industrial applications, International Centre for Mathematical Sciences, £900 for Summer 2026

ILAS 2026 Early Career NSF Conference Grant,

International Linear Algebra Society, Virginia Tech, \$400 for Spring 2026

Conference Travel Grant,

Institute of Computational and Experimental Research in Mathematics, Randomized Numerical Linear Algebra, \$850 for Spring 2026

Conference Travel Grant,

Carleton College, Science Education Resource Center at Carleton College, \$500 for Fall 2025

Summer Student Showcase Best Presenter,

Lawrence Livermore National Laboratory, Data Science Institute, 2025.

Conference Travel Grant,

Southeast Applied and Computational Mathematics Student Conference, Auburn University, \$200 for Spring 2025

Summer Research Presentation Superlative,

Sustainable Horizons Institute, 2024.

Piedmont Teaching Fellowship in Sustainability and Curriculum Development,

Emory University, \$1,000 for AY 2024 - 2025, 2025 - 2026.

Conference Travel Grant,

Institute of Computational and Experimental Research in Mathematics, Acceleration and Extrapolation Methods, \$850 for Summer 2023

Research Conference Presenter Grant,

Tufts University, Graduate School of Arts and Science, \$600 for AY 2022 - 2023

Department of Mathematics Scholarship,

Tufts University, \$20,000 for AY 2022 - 2023

The Fuertes Medal Memorial Prize for Public Speaking,

Cornell University, College of Engineering, \$3,000 for AY 2019 - 2020.

MEMBERSHIPS

American Mathematical Society (AMS)

Society for Industrial and Applied Mathematics (SIAM)

Spectral!

SERVICE

Conference Organization:

Organizer, Randomized, iterative algorithms for data science, SIAM Conference on Mathematics of Data Science (MDS26), November 16-17, 2026

Conference Reviewer: [SIAM International Conference on Data Mining \(SDM26\)](#), November 19-20, 2026, Salt Lake City, Utah.

Local Organizing Committee: [Southeast Applied & Computational Math](#), April 3-4, 2026, Emory University, Atlanta, GA.

DEPARTMENTAL SERVICE

Emory University

Officer: Directed Reading Program Steering Committee, 2023 - Present

President: SIAM Chapter, 2026 - Present

Risk Manager: SIAM Chapter, 2025 - 2026

Site Coordinator: American Mathematics Competition (AMC-8, American Invitational Mathematics Examination), 2025 - Present

STEM
OUTREACH

Atlanta Science Festival Exploration Expo Exhibitor, "Math Makes it Happen!", March 22, 2025

Mary Lin Elementary School's STEAM Night, November 7th, 2024

SOFTWARE
SKILLS

English: native speaker

Programming: MATLAB, JULIA, PYTHON, C/C++

Computer: L^AT_EX, Microsoft Office, HTML, CSS

LANGUAGE
SKILLS

English: native speaker

Programming: MATLAB, JULIA, PYTHON, C/C++

Computer: L^AT_EX, Microsoft Office, HTML, CSS